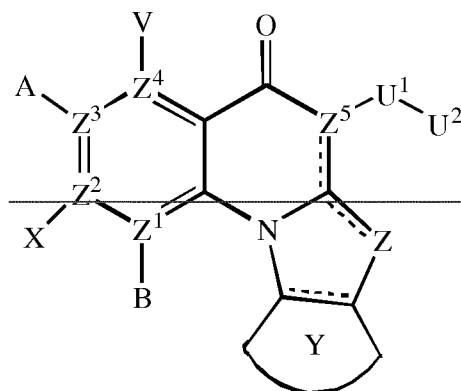


Amended claims (marked-up version)

1. ~~_____ A compound of Formula (I),~~



Formula (I)

~~or a pharmaceutically acceptable salt or ester thereof;~~

~~wherein _____ indicates an optionally unsaturated bond;~~

~~each of B, X, A or V is absent if Z¹, Z², Z³ and Z⁴, respectively, is N; and~~

~~each of B, X, A and V is independently H, halo, azido, _____ CN, _____ CF₃, _____~~

~~CONR¹R², _____ NR¹R², SR², OR², R³, W, _____ L W, _____ W⁰, _____ L N(R) _____ W⁰, A² or A³, when~~
~~each of Z¹, Z², Z³ and Z⁴, respectively, is C;~~

~~Z is O, S, CR⁴₂, NR⁴CR⁴, CR⁴NR⁴, CR⁴, NR⁴ or N;~~

~~each of Z¹, Z², Z³ and Z⁴ is independently C or N, provided any three N are non-~~
~~adjacent;~~

~~Z⁵ is C; or Z⁵ may be N when Z is N;~~

~~Y is an optionally substituted 5-6 membered carbocyclic or heterocyclic ring;~~

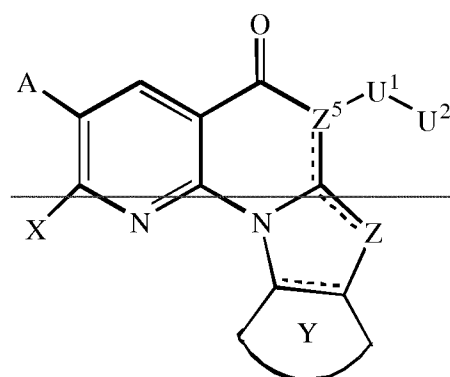
~~U¹ is C(____T)N(R), C(____T)N(R)O, C(____T), _____ SO₂N(R), _____ SO₂N(R)N(R⁰), _____ SO₂, or~~
~~_____ SO₃, where T is O, S, or NH; or U¹ may be a bond when Z⁵ is N or U² is H;~~

~~U² is H, or C3-C7 cycloalkyl, C1-C10 alkyl, C1-C10 heteroalkyl, C2-C10 alkenyl~~
~~or C2-C10 heteroalkenyl group, each of which may be optionally substituted~~

~~with one or more halogens, =O, or an optionally substituted 3-7 membered carbocyclic or heterocyclic ring; or U² is W, L-W, L-N(R)-W⁰, A² or A³; in each NR¹R², R¹ and R² together with N may form an optionally substituted azaicyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member;~~
 5 ~~R¹ is H or C1-C6 alkyl, optionally substituted with one or more halogens, or =O; R² is H, or C1-C10 alkyl, C1-C10 heteroalkyl, C2-C10 alkenyl, or C2-C10 heteroalkenyl, each of which may be optionally substituted with one or more~~
 10 ~~halogens, =O, or an optionally substituted 3-7 membered carbocyclic or heterocyclic ring; R³ is an optionally substituted C1-C10 alkyl, C2-C10 alkenyl, C5-C10 aryl, or C6-C12 arylalkyl, or a heteroform of one of these, each of which may be optionally substituted with one or more halogens, =O, or an optionally~~
 15 ~~substituted 3-6 membered carbocyclic or heterocyclic ring; each R⁴ is independently H, or C1-C6 alkyl; or R⁴ may be W, L-W or L-N(R)-W⁰; each R and R⁰ is independently H or C1-C6 alkyl; L is a C1-C10 alkylene, C1-C10 heteroalkylene, C2-C10 alkenylene or C2-C10~~
 20 ~~heteroalkenylene linker, each of which may be optionally substituted with one or more substituents selected from the group consisting of halogen, oxo (=O), or C1-C6 alkyl; W is an optionally substituted 4-7 membered azaicyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring~~
 25 ~~member; W⁰ is an optionally substituted 3-4 membered carbocyclic ring, or a C1-C6 alkyl group substituted with from 1 to 4 fluorine atoms; provided one of U², V, A, X and B is a secondary amine A² or a tertiary amine A³, wherein~~
 30 ~~the secondary amine A² is NH-W⁰, and~~

the tertiary amine A^3 is a fully saturated and optionally substituted six-membered or seven-membered azacyclic ring optionally containing an additional heteroatom selected from N, O or S as a ring member, or the tertiary amine A^3 is a partially unsaturated or aromatic optionally substituted five-membered azacyclic ring, optionally containing an additional heteroatom selected from N, O or S as a ring member;
for use in treating an autoimmune disease in a subject.

2. A compound of Formula (II),



Formula (II)

or a pharmaceutically acceptable salt or ester thereof;

wherein --- indicates an optionally unsaturated bond;

each of A and X is independently H, halo, azido, CN, CF_3 , CONR^1R^2 , NR^1R^2 , SR^2 , OR^2 , R^3 , W, L W, W^0 , L N(R) W^0 , A^2 or A^3 ;

Z is O, S, CR^4_2 , NR^4CR^4 , CR^4NR^4 or NR^4 ;

Y is an optionally substituted 5-6 membered carbocyclic or heterocyclic ring;

U^1 is $\text{C}(-\text{T})\text{N}(\text{R})$, $\text{C}(-\text{T})\text{N}(\text{R})\text{O}$, $\text{C}(-\text{T})$, $\text{SO}_2\text{N}(\text{R})$, $\text{SO}_2\text{N}(\text{R})\text{N}(\text{R}^0)$, SO_2 , or SO_3 , where T is O, S, or NH; or U^1 may be a bond when U^2 is H;

U^2 is H, or C3-C7 cycloalkyl, C1-C10 alkyl, C1-C10 heteroalkyl, C2-C10 alkenyl or C2-C10 heteroalkenyl group, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-7 membered carbocyclic or heterocyclic ring; or U^2 is W, L W, L N(R) W^0 , A^2 or A^3 ;

~~in each NR^1R^2 , R^1 and R^2 together with N may form an optionally substituted azaacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member;~~

~~R^1 is H or C1-C6 alkyl, optionally substituted with one or more halogens, or =O;~~

~~R^2 is H, or C1-C10 alkyl, C1-C10 heteroalkyl, C2-C10 alkenyl, or C2-C10 heteroalkenyl, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-7 membered carbocyclic or heterocyclic ring;~~

~~R^3 is an optionally substituted C1-C10 alkyl, C2-C10 alkenyl, C5-C10 aryl, or C6-C12 arylalkyl, or a heteroform of one of these, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-6 membered carbocyclic or heterocyclic ring;~~

~~each R^4 is independently H, or C1-C6 alkyl; or R^4 may be W, L-W or L-N(R)-W;~~

~~each R and R^0 is independently H or C1-C6 alkyl;~~

~~L is a C1-C10 alkylene, C1-C10 heteroalkylene, C2-C10 alkenylene or C2-C10 heteroalkenylene linker, each of which may be optionally substituted with one or more substituents selected from the group consisting of halogen, oxo (=O), or C1-C6 alkyl;~~

~~W is an optionally substituted 4-7 membered azaacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member;~~

~~W^0 is an optionally substituted 3-4 membered carbocyclic ring, or a C1-C6 alkyl group substituted with from 1 to 4 fluorine atoms;~~

~~provided that one of U^2 , A, and X is a secondary amine A^2 or a tertiary amine A^3 , wherein~~

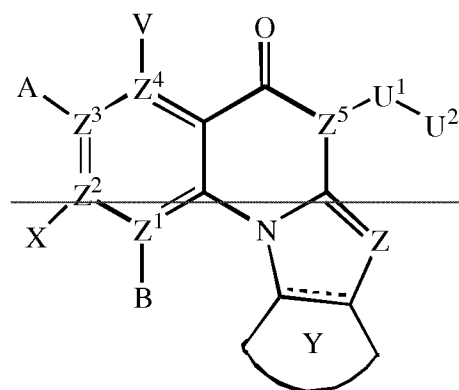
~~the secondary amine A^2 is NH-W^0 , and~~

~~the tertiary amine A^3 is a fully saturated and optionally substituted six-membered or seven-membered azaacyclic ring optionally containing an~~

additional heteroatom selected from N, O or S as a ring member, or the tertiary amine A³ is a partially unsaturated or aromatic optionally substituted five-membered azaicyclic ring optionally containing an additional heteroatom selected from N, O or S as a ring member;

5 for use in treating an autoimmune disease in a subject.

3. ——— A compound of Formula (III),



Formula (III)

10

or a pharmaceutically acceptable salt or ester thereof;

wherein — indicates an optionally unsaturated bond; and

each of B, X, A or V is absent if Z¹, Z², Z³ and Z⁴, respectively, is N; and

each of B, X, A and V is independently H, halo, azido, CN, CF₃, CONR¹R², —

15

NR¹R², SR², OR², R³, W, L W, W⁰, L N(R) W⁰, A² or A³, when each of Z¹, Z², Z³ and Z⁴, respectively, is C;

each of Z¹, Z², Z³ and Z⁴ is independently C or N, provided any three N are non-adjacent;

Y is an optionally substituted 5-6 membered carbocyclic or heterocyclic ring;

20

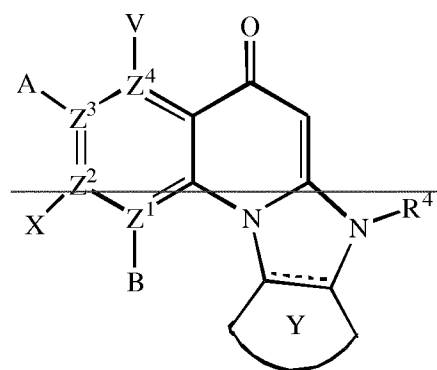
U¹ is C(=T)N(R), C(=T)N(R)O, C(=T), SO₂N(R), SO₂N(R)N(R⁰), SO₂, or —SO₃, where T is O, S, or NH; or U¹ may be a bond when Z⁵ is N or U² is H;

U² is H, or C₃-C₇ cycloalkyl, C₁-C₁₀ alkyl, C₁-C₁₀ heteroalkyl, C₂-C₁₀ alkenyl or C₂-C₁₀ heteroalkenyl group, each of which may be optionally substituted

~~with one or more halogens, =O, or an optionally substituted 3-7 membered carbocyclic or heterocyclic ring; or U² is -W, -L-W, -L-N(R)-W⁰, A² or A³; in each -NR¹R², R¹ and R² together with N may form an optionally substituted azaicyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member;~~
~~R¹ is H or C1-C6 alkyl, optionally substituted with one or more halogens, or =O;~~
~~R² is H, or C1-C10 alkyl, C1-C10 heteroalkyl, C2-C10 alkenyl, or C2-C10 heteroalkenyl, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-7 membered carbocyclic or heterocyclic ring;~~
~~R³ is an optionally substituted C1-C10 alkyl, C2-C10 alkenyl, C5-C10 aryl, or C6-C12 arylalkyl, or a heteroform of one of these, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-6 membered carbocyclic or heterocyclic ring;~~
~~each R⁴ is independently H, or C1-C6 alkyl; or R⁴ may be -W, -L-W or -L-N(R)-W⁰;~~
~~each R and R⁰ is independently H or C1-C6 alkyl;~~
~~L is a C1-C10 alkylene, C1-C10 heteroalkylene, C2-C10 alkenylene or C2-C10 heteroalkenylene linker, each of which may be optionally substituted with one or more substituents selected from the group consisting of halogen, oxo (=O), or C1-C6 alkyl;~~
~~W is an optionally substituted 4-7 membered azaicyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member;~~
~~W⁰ is an optionally substituted 3-4 membered carbocyclic ring, or a C1-C6 alkyl group substituted with from 1 to 4 fluorine atoms;~~
~~provided that one of U², V, A, X and B is a secondary amine A² or a tertiary amine A³, wherein~~
~~the secondary amine A² is -NH-W⁰, and~~

the tertiary amine A^3 is a fully saturated and optionally substituted six-membered or seven-membered azacyclic ring optionally containing an additional heteroatom selected from N, O or S as a ring member, or the tertiary amine A^3 is a partially unsaturated or aromatic optionally substituted five-membered azacyclic ring, optionally containing an additional heteroatom selected from N, O or S as a ring member, for use in treating an autoimmune disease in a subject.

4. A compound of Formula (IV),



Formula (IV)

or a pharmaceutically acceptable salt or ester thereof;

wherein --- indicates an optionally unsaturated bond;

each of B, X, A or V is absent if Z^1 , Z^2 , Z^3 and Z^4 respectively, is N; and

each of B, X, A and V is independently H, halo, azido, CN, CF_3 , CONR^1R^2 , NR^1R^2 , SR^2 , OR^2 , R^3 , W, L W, W^0 , L N(R) W^0 , A^2 or A^3 , when each of Z^1 , Z^2 , Z^3 and Z^4 , respectively, is C;

each of Z^1 , Z^2 , Z^3 and Z^4 is independently C or N, provided any three N are non-adjacent;

Y is an optionally substituted 5-6 membered carbocyclic or heterocyclic ring;

in each NR^1R^2 , R^1 and R^2 together with N may form an optionally substituted azacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member;

~~R¹ is H or C1-C6 alkyl, optionally substituted with one or more halogens, or =O;~~

~~R² is H, or C1-C10 alkyl, C1-C10 heteroalkyl, C2-C10 alkenyl, or C2-C10 heteroalkenyl, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-7 membered carbocyclic or heterocyclic ring;~~

~~R³ is an optionally substituted C1-C10 alkyl, C2-C10 alkenyl, C5-C10 aryl, or C6-C12 arylalkyl, or a heteroform of one of these, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-6 membered carbocyclic or heterocyclic ring;~~

~~R⁴ is W, L-W or L-N(R)-W⁰; and~~

~~each R is independently H or C1-C6 alkyl;~~

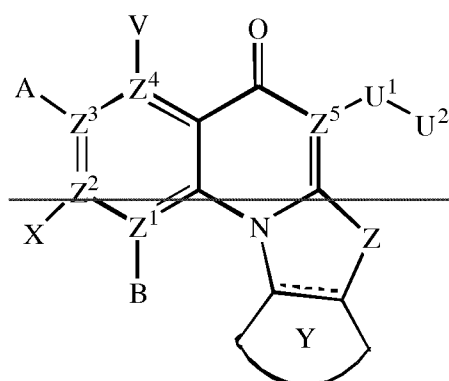
~~L is a C1-C10 alkylene, C1-C10 heteroalkylene, C2-C10 alkenylene or C2-C10 heteroalkenylene linker, each of which may be optionally substituted with one or more substituents selected from the group consisting of halogen, oxo (=O), or C1-C6 alkyl;~~

~~W is an optionally substituted 4-7 membered azacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member; and~~

~~W⁰ is an optionally substituted 3-4 membered carbocyclic ring, or a C1-C6 alkyl group substituted with from 1 to 4 fluorine atoms,~~

~~for use in treating an autoimmune disease in a subject.~~

5. ~~— A compound of Formula (V),~~



Formula (V)

5 ~~or a pharmaceutically acceptable salt or ester thereof;~~

~~— indicates an optionally unsaturated bond;~~

~~A and V independently are H, halo, azido, CN, CF₃, CONR¹R², NR¹R², SR², OR², R³, W, L W, W⁰, L N(R) W⁰, A² or A³;~~

~~Z is O, S, CR⁴₂, NR⁴CR⁴, CR⁴NR⁴ or NR⁴;~~

10 ~~Y is an optionally substituted 5-6 membered carbocyclic or heterocyclic ring;~~

~~U¹ is C(=T)N(R), C(=T)N(R)O, C(=T), SO₂N(R), SO₂N(R)N(R⁰), SO₂, or SO₃, where T is O, S, or NH; or U¹ may be a bond when U² is H;~~

~~U² is H, or C3-C7 cycloalkyl, C1-C10 alkyl, C1-C10 heteroalkyl, C2-C10 alkenyl or C2-C10 heteroalkenyl group, each of which may be optionally substituted~~

15 ~~with one or more halogens, =O, or an optionally substituted 3-7 membered carbocyclic or heterocyclic ring; or U² is W, L W or L N(R) W⁰, A² or A³;~~

~~in each NR¹R², R¹ and R² together with N may form an optionally substituted azaicyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member;~~

20 ~~R¹ is H or C1-C6 alkyl, optionally substituted with one or more halogens, or =O;~~

~~R² is H, or C1-C10 alkyl, C1-C10 heteroalkyl, C2-C10 alkenyl, or C2-C10 heteroalkenyl, each of which may be optionally substituted with one or more~~

halogens, =O, or an optionally substituted 3-7 membered carbocyclic or heterocyclic ring;

R^3 is an optionally substituted C1-C10 alkyl, C2-C10 alkenyl, C5-C10 aryl, or C6-C12 arylalkyl, or a heteroform of one of these, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-6 membered carbocyclic or heterocyclic ring;

each R^4 is independently H, or C1-C6 alkyl; or R^4 may be W , $L-W$ or $L-N(R)$ - W^0 ;

each R and R^0 is independently H or C1-C6 alkyl;

L is a C1-C10 alkylene, C1-C10 heteroalkylene, C2-C10 alkenylene or C2-C10 heteroalkenylene linker, each of which may be optionally substituted with one or more substituents selected from the group consisting of halogen, oxo (=O), or C1-C6 alkyl;

W is an optionally substituted 4-7 membered azacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member;

W^0 is an optionally substituted 3-4 membered carbocyclic ring, or a C1-C6 alkyl group substituted with from 1 to 4 fluorine atoms;

provided one of U^2 , A , and V is a secondary amine A^2 or a tertiary amine A^3 ,

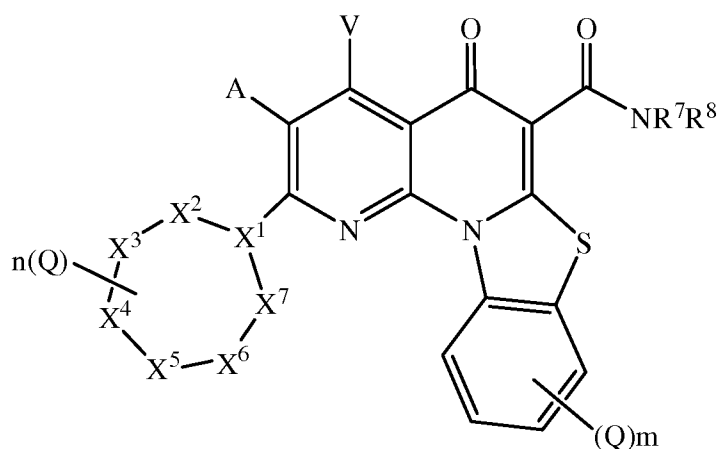
wherein

the secondary amine A^2 is $NH-W^0$, and

the tertiary amine A^3 is a fully saturated and optionally substituted six-membered or seven-membered azacyclic ring optionally containing an additional heteroatom selected from N, O or S as a ring member, or the tertiary amine A^3 is a partially unsaturated or aromatic optionally substituted five-membered azacyclic ring optionally containing an additional heteroatom selected from N, O or S as a ring member,

for use in treating an autoimmune disease in a subject.

61. A compound of Formula (VI),



Formula (VI)

5 or a pharmaceutically acceptable salt or ester thereof;

wherein:

X¹ is CH or N;

X², X³, X⁴, X⁵, X⁶ and X⁷ independently are NR⁴, CH₂, CHQ or C(Q)₂, provided
 that: (i) zero, one or two of X², X³, X⁴, X⁵, X⁶ and X⁷ are NR⁴; (ii) when X¹ is N,
 10 both of X² and X⁷ are not NR⁴; (iii) when X¹ is N, X³ and X⁶ are not NR⁴; and
 (iv) when X¹ is CH and two of X², X³, X⁴, X⁵, X⁶ and X⁷ are NR⁴, the two NR⁴ are
 located at adjacent ring positions or are separated by two or more other ring
 positions;

A and V independently are H, halo, azido, -CN, -CF₃, -CONR¹R², -NR¹R², -SR²,
 15 -OR², -R³, -W, -L-W, -W⁰, or -L-N(R)-W⁰;

each Q is independently halo, azido, -CN, -CF₃, -CONR¹R², -NR¹R², -SR², -OR²,
 -R³, -W, -L-W, -W⁰, or -L-N(R)-W⁰;

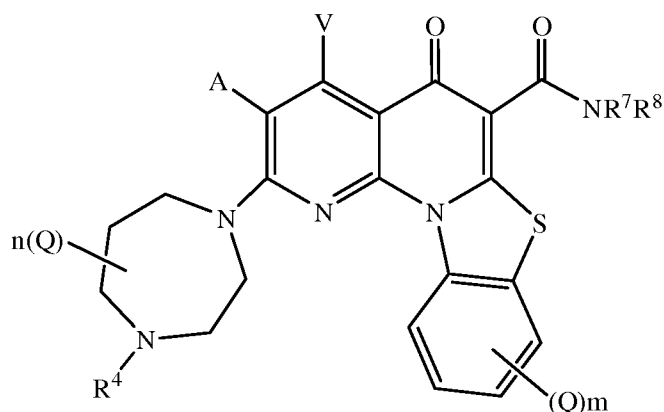
in each —NR¹R², R¹ and R² together with N may form an optionally substituted
 azacyclic ring, optionally containing one additional heteroatom selected from N,

20 O and S as a ring member;

R¹ is H or C1-C6 alkyl, optionally substituted with one or more halogens, or
 =O;

- R is H, or C1-C10 alkyl, C1-C10 heteroalkyl, C2-C10 alkenyl, or C2-C10 heteroalkenyl, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-7 membered carbocyclic or heterocyclic ring;
- 5 R³ is an optionally substituted C1-C10 alkyl, C2-C10 alkenyl, C5-C10 aryl, or C6-C12 arylalkyl, or a heteroform of one of these, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-6 membered carbocyclic or heterocyclic ring;
- each R⁴ is independently H, or C1-C6 alkyl; or R⁴ may be -W, -L-W or -L-N(R)-
- 10 W⁰;
- each R is independently H or C1-C6 alkyl;
- R⁷ is H and R⁸ is C1-C10 alkyl, C1-C10 heteroalkyl, C2-C10 alkenyl, or C2-C10 heteroalkenyl, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-7 membered carbocyclic or
- 15 heterocyclic ring; or in —NR⁷R⁸, R⁷ and R⁸ together with N may form an optionally substituted azacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member;
- m is 0, 1, 2, 3 or 4;
- n is 0, 1, 2, 3, 4, or 5;
- 20 L is a C1-C10 alkylene, C1-C10 heteroalkylene, C2-C10 alkenylene or C2-C10 heteroalkenylene linker, each of which may be optionally substituted with one or more substituents selected from the group consisting of halogen, oxo (=O), or C1-C6 alkyl;
- W is an optionally substituted 4-7 membered azacyclic ring, optionally
- 25 containing an additional heteroatom selected from N, O and S as a ring member; and
- W⁰ is an optionally substituted 3-4 membered carbocyclic ring, or a C1-C6 alkyl group substituted with from 1 to 4 fluorine atoms,
- for use in treating ~~an autoimmune disease~~ multiple sclerosis in a subject.

72. A compound of Formula (VIII),



Formula (VIII)

5 or a pharmaceutically acceptable salt or ester thereof;
wherein:

A and V independently are H, halo, azido, -CN, -CF₃, -CONR¹R², -NR¹R², -SR²,
-OR², -R³, -W, -L-W, -W⁰, or -L-N(R)-W⁰;

each Q is independently halo, azido, -CN, -CF₃, -CONR¹R², -NR¹R², -SR², -OR²,
10 -R³, -W, -L-W, -W⁰, or -L-N(R)-W⁰;

in each —NR¹R², R¹ and R² together with N may form an optionally substituted
azacyclic ring, optionally containing an additional heteroatom selected from N,
O and S as a ring member;

R¹ is H or C1-C6 alkyl, optionally substituted with one or more halogens, or
15 =O;

R² is H, or C1-C10 alkyl, C1-C10 heteroalkyl, C2-C10 alkenyl, or C2-C10
heteroalkenyl, each of which may be optionally substituted with one or more
halogens, =O, or an optionally substituted 3-7 membered carbocyclic or
heterocyclic ring;

20 R³ is an optionally substituted C1-C10 alkyl, C2-C10 alkenyl, C5-C10 aryl, or
C6-C12 arylalkyl, or a heteroform of one of these, each of which may be
optionally substituted with one or more halogens, =O, or an optionally
substituted 3-6 membered carbocyclic or heterocyclic ring;

each R⁴ is independently H, or C1-C6 alkyl; or R⁴ may be -W, -L-W or -L-N(R)-W⁰;

each R is independently H or C1-C6 alkyl;

R⁷ is H and R⁸ is C1-C10 alkyl, C1-C10 heteroalkyl, C2-C10 alkenyl, or C2-C10

5 heteroalkenyl, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-7 membered carbocyclic or heterocyclic ring; or in —NR⁷R⁸, R⁷ and R⁸ together with N may form an optionally substituted azacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member;

10 m is 0, 1, 2, 3 or 4;

n is 0, 1, 2, 3, 4 or 5;

p is 0, 1, 2 or 3;

L is a C1-C10 alkylene, C1-C10 heteroalkylene, C2-C10 alkenylene or C2-C10 heteroalkenylene linker, each of which may be optionally substituted with one

15 or more substituents selected from the group consisting of halogen, oxo (=O), or C1-C6 alkyl;

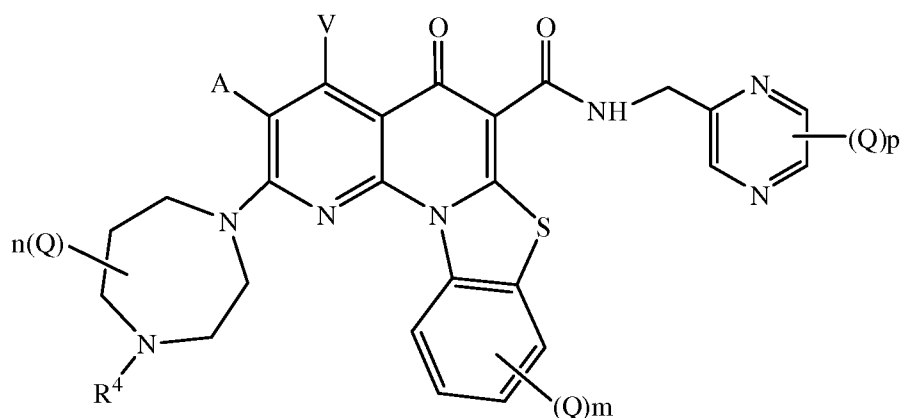
W is an optionally substituted 4-7 membered azacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member; and

20 W⁰ is an optionally substituted 3-4 membered carbocyclic ring, or a C1-C6 alkyl group substituted with from 1 to 4 fluorine atoms,

for use in treating ~~an autoimmune disease~~ multiple sclerosis in a subject.

25 3. A compound for use according to claim 1 or 2, wherein R⁷ is H and R⁸ is a C₁₋₄ alkyl substituted with an optionally substituted aromatic heterocyclic ring.

30 4. A compound for use according to claim 3, wherein the optionally substituted aromatic heterocyclic ring is selected from pyridine, pyrimidine, pyrazine, imidazole, pyrrolidine, and thiazole.



Formula (VII)

- 5 or a pharmaceutically acceptable salt or ester thereof;
A and V independently are H, halo, azido, -CN, -CF₃, -CONR¹R², -NR¹R², -SR², -OR², -R³, -W, -L-W, -W⁰, or -L-N(R)-W⁰;
each Q is independently halo, azido, -CN, -CF₃, -CONR¹R², -NR¹R², -SR², -OR², -R³, -W, -L-W, -W⁰, or -L-N(R)-W⁰;
10 in each —NR¹R², R¹ and R² together with N may form an optionally substituted azacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member;
R¹ is H or C1-C6 alkyl, optionally substituted with one or more halogens, or =O;
15 R² is H, or C1-C10 alkyl, C1-C10 heteroalkyl, C2-C10 alkenyl, or C2-C10 heteroalkenyl, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-7 membered carbocyclic or heterocyclic ring;
R³ is an optionally substituted C1-C10 alkyl, C2-C10 alkenyl, C5-C10 aryl, or
20 C6-C12 arylalkyl, or a heteroform of one of these, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-6 membered carbocyclic or heterocyclic ring;

each R⁴ is independently H, or C1-C6 alkyl; or R⁴ may be -W, -L-W or -L-N(R)-W⁰;

each R is independently H or C1-C6 alkyl;

m is 0, 1, 2, 3 or 4;

5 n is 0, 1, 2, 3, 4 or 5;

p is 0, 1, 2 or 3;

L is a C1-C10 alkylene, C1-C10 heteroalkylene, C2-C10 alkenylene or C2-C10 heteroalkenylene linker, each of which may be optionally substituted with one or more substituents selected from the group consisting of halogen, oxo (=O), or

10 C1-C6 alkyl;

W is an optionally substituted 4-7 membered azacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member; and

15 W⁰ is an optionally substituted 3-4 membered carbocyclic ring, or a C1-C6 alkyl group substituted with from 1 to 4 fluorine atoms,

for use in treating ~~an autoimmune disease~~ multiple sclerosis in a subject.

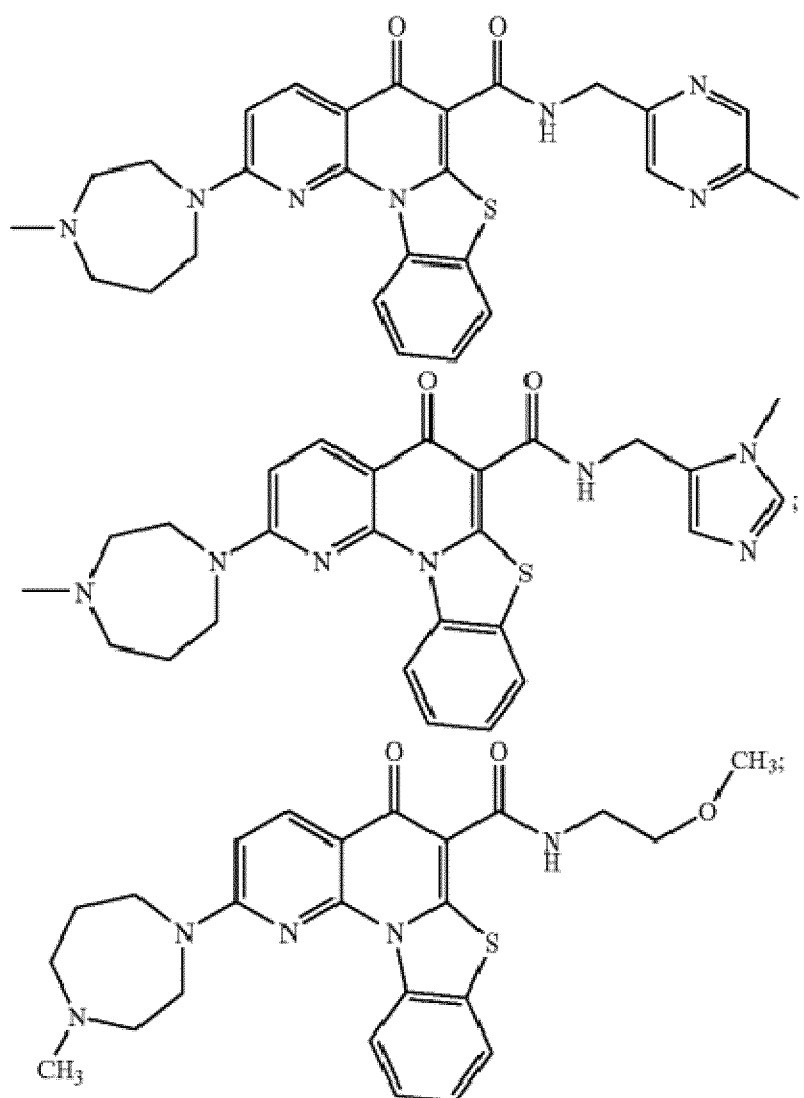
6. A compound for use according to claim 5, wherein p is 0 or 1.

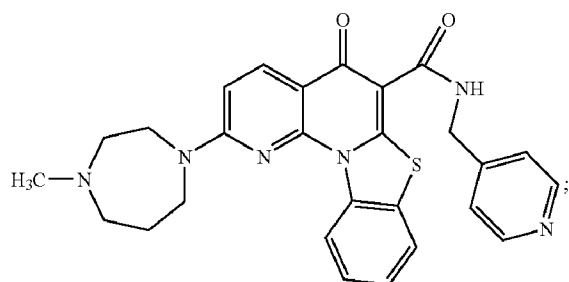
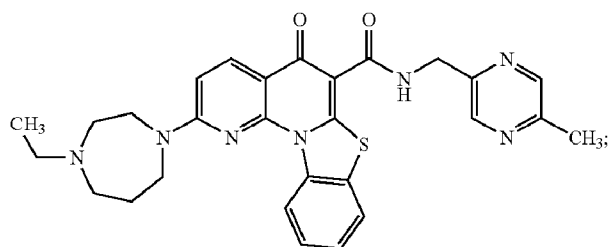
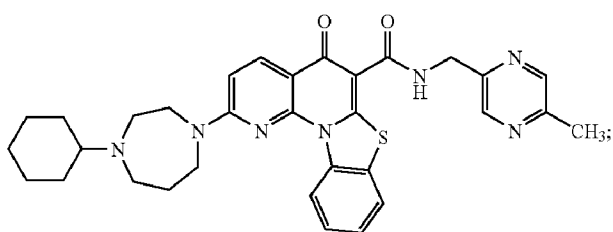
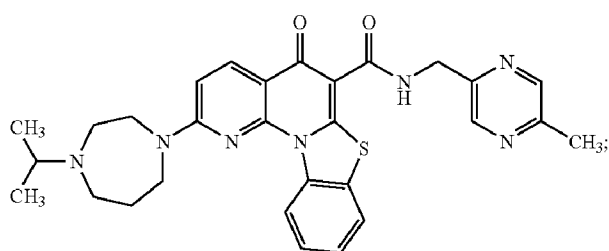
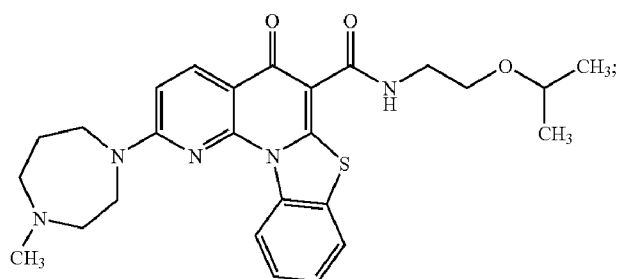
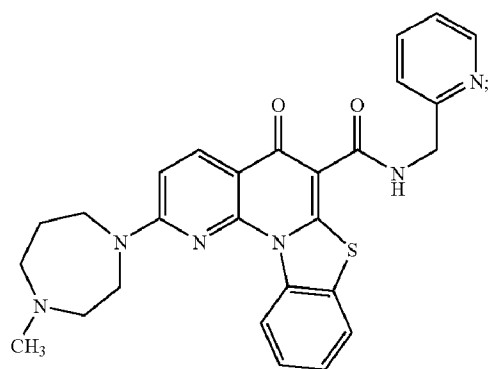
20 7. A compound for use according to any of claims 1 to 6, wherein A and V are independently H or halo.

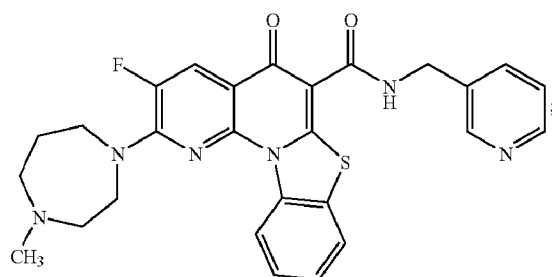
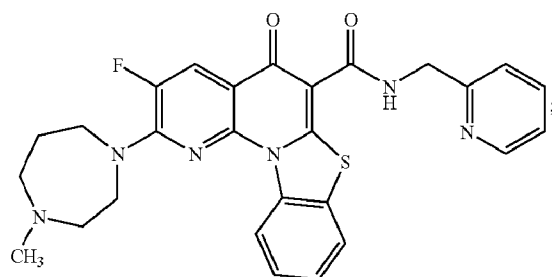
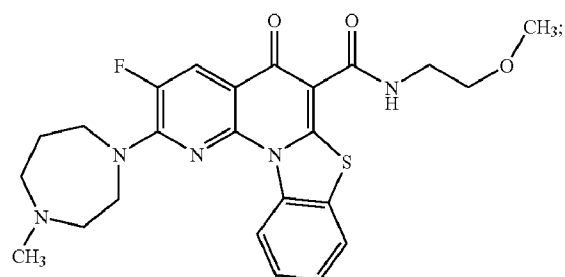
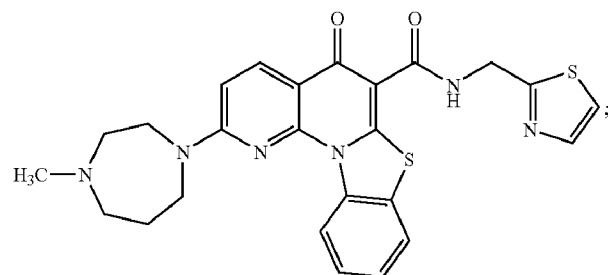
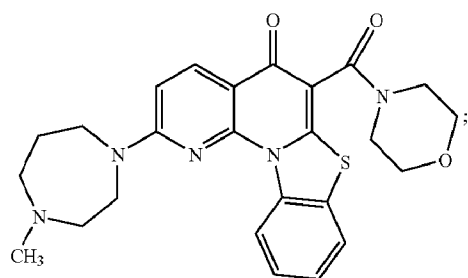
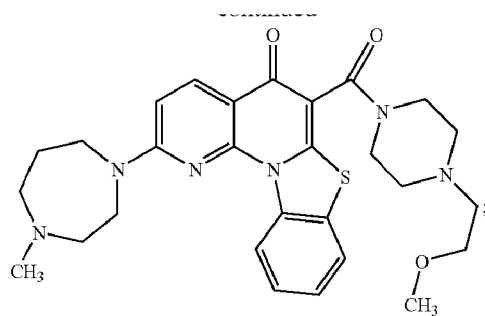
8. A compound for use according to any of claims 1 to 7, wherein R⁴ is H or C1-4 alkyl.

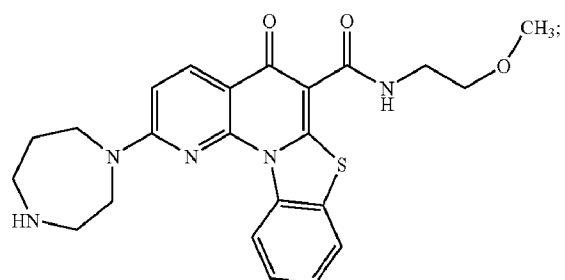
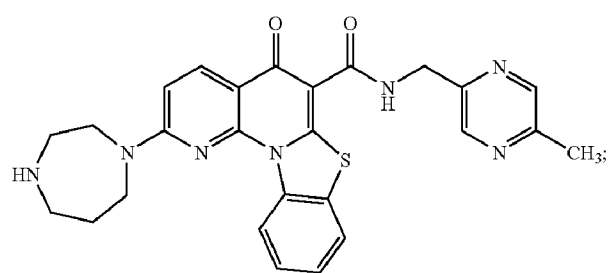
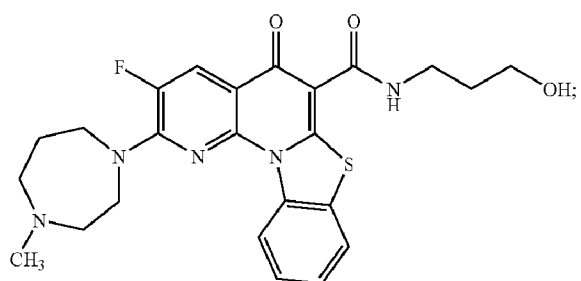
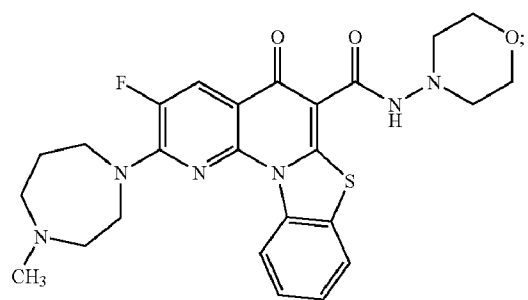
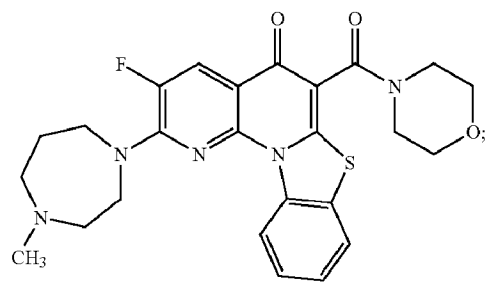
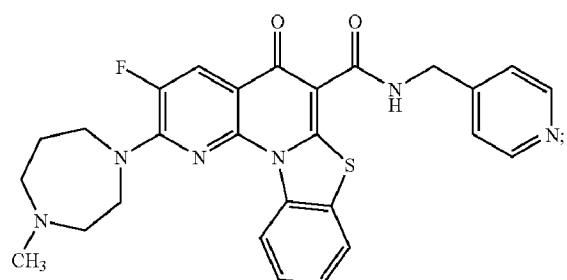
25 9. A compound for use according to any of claims 1 to 8, wherein m and n are each 0.

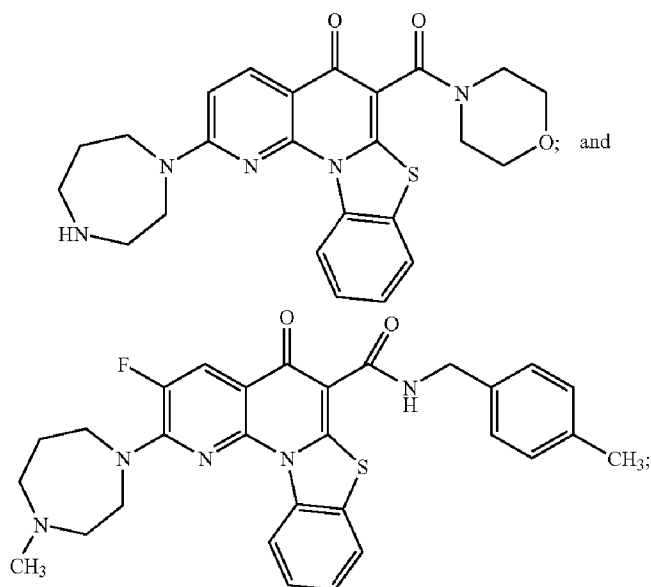
910. A compound selected from the group consisting of:











or a pharmaceutically acceptable salt or ester thereof,

for use in treating ~~an autoimmune disease~~ multiple sclerosis in a subject.

~~1011.~~ A method of monitoring treatment efficiency of a subject having ~~an autoimmune disease~~ multiple sclerosis, the method comprising:

comparing a level of expression of least one gene involved in the RNA polymerase I pathway in a cell of the subject who has been treated with said compound of any one of claims 1-9-10 to a level of expression of said at least one gene in a cell of the subject prior to being treated with said compound,

(i) wherein a decrease above a predetermined threshold in said level of expression of said at least one gene following said treating with said compound relative to said level of expression of said at least one gene prior to said treating with said compound indicates that said compound is efficient for treating the subject;

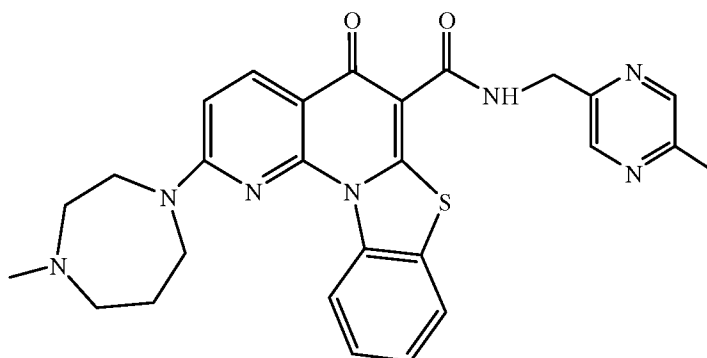
(ii) wherein an increase above a predetermined threshold in said level of expression of said at least one gene following said treating with said compound relative to said level of expression of said at least one gene prior to

said treating with said compound indicates that said compound is not efficient for treating the subject; or

(iii) wherein when a level of expression of said at least one gene following said treating with said compound is identical or changed below a predetermined threshold as compared to prior to said treating with said compound then the treatment is not efficient for treating the subject

thereby monitoring treatment efficiency of the subject having an ~~autoimmune disease~~ multiple sclerosis.

~~11~~12. A compound for use according to any one of claims 1-~~9~~10, or the method of claim ~~10~~11, wherein said compound is 2-(4-methyl-1,4-diazepan-1-yl)-N-((5-methylpyrazin-2-yl)methyl)-5-oxo-5H-benzo[4,5]thiazolo[3,2-a][1,8]naphthyridine-6-carboxamide as depicted in Formula



~~12. The use of any one of claims 1-9 and 11, or the method of claim 10 or 11, wherein the autoimmune disease is multiple sclerosis.~~

13. A compound for use according to any of claims 1-10 or the method of claim 11 or 12, wherein said multiple sclerosis is a relapsing-remitting course of multiple sclerosis.

14. A compound for use according to any of claims 1-10 or the method of claim 11 or 12, wherein said multiple sclerosis is a progressive course of multiple sclerosis.

15. A compound for use according to any of claims 1-10 or the method of claim 11 or 12, wherein said multiple sclerosis is a benign multiple sclerosis.